MID-STATE TECHNICAL COLLEGE · IT PROGRAMS

High Availability and Enterprise Features

Unit 12 · Virtualization and SDDC (10-150-130)

MID-STATE TECHNICAL COLLEGE

Learning Outcomes

By the end of this unit, you will be able to:



Build and manage a cluster

Create a three-node Proxmox VE cluster on a dedicated network and explain how quorum protects it.



Implement high availability

Migrate guests, schedule replication, and configure HA resources that fail over automatically.



Explain distributed storage

Describe Ceph's RADOS, CRUSH, and OSD model, and compare replication with erasure coding.



Plan for enterprise scale

Apply RTO/RPO, N+1 capacity, and active-active vs. active-passive patterns to a design.

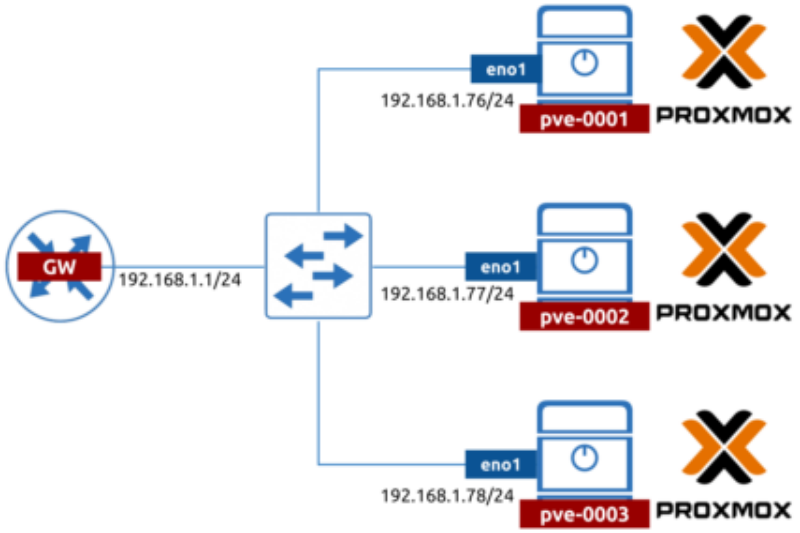
LESSON 1

Clustering and Quorum



Why Cluster?

- A **cluster** joins independent hosts into one management domain — log in to any node and manage them all.
- Configuration lives in the **Proxmox Cluster File System (pmxcfs)**, mounted at `/etc/pve` and replicated to every node in real time.
- **Corosync** carries cluster membership and messaging; it decides who is in the cluster and who is not.
- Clustering unlocks **migration**, **replication**, and **high availability** — but does not provide them by itself.



Three hosts, one switch, one management plane.

Cluster Building Blocks

Four components turn three servers into one cluster:

Component	What it does	Where you see it
Corosync	Membership, messaging, and voting between nodes over the cluster link	`/etc/pve/corosync.conf`
pmxcfs	Database-backed file system that replicates config to every node	`/etc/pve` on each host
pvecm	Command-line tool to create, join, and inspect the cluster	`pvecm status`, `pvecm nodes`
Kronosnet links	Up to eight redundant corosync links for failover of the cluster network	Join dialog → link 0, link 1

Corosync wants low, stable latency — keep it on a dedicated network and off WAN links.

Quorum: Majority Rules

A partition may only make changes if it holds more than half of the votes.

Nodes	Votes needed	Failures tolerated	Notes
2	2	0	Needs a QDevice to be useful
3	2	1	Lab 14 pods — the minimum for real HA
4	3	1	Same tolerance as 3; a 2/2 split stops both halves
5	3	2	Common production starting point

Formula: `quorum = floor(nodes ÷ 2) + 1`. Without quorum, /etc/pve goes read-only and nothing can start or move.

Split-Brain and the QDevice

- **Split-brain**: a network partition leaves two groups of nodes that each think they are in charge.
- If both sides start the same VM against shared storage, the disk is **corrupted** — this is the outcome quorum exists to prevent.
- A **QDevice** (`corosync-qnetd` on an outside host) adds one tie-breaking vote over TCP.
- Use it for **even** node counts, especially two-node clusters; avoid it on odd counts.



Do not “force” quorum

`pvecm expected 1` lets a lone node act without a majority. It is a recovery tool for a dead cluster — never a fix for a flapping one.

Creating the Cluster

Lab 14 builds Cluster1 on the dedicated 10.0.200.x/24 cluster network:

```
# Host1 — create the cluster, binding corosync to the cluster network
pvecm create Cluster1 --link0 10.0.200.1
# Host2 / Host3 — join through Host1, using each node's own cluster IP
pvecm add 10.0.200.1 --link0 10.0.200.2
pvecm status        # Quorate: Yes · Total votes: 3
```

- Joining nodes must be empty — no VMs or containers — and on the same major version.
- GUI path: **Datacenter → Cluster → Create Cluster**, then **Join Information** on the new members.

LESSON 2

Migration, Replication, and HA



Offline vs. Live Migration

Offline migration

- Guest is **shut down**, moved, then started on the target.
- Downtime = shutdown + copy + boot.
- Works for VMs and containers; containers use **restart mode**.
- `qm migrate 101 Host2`

Live migration

- Guest **keeps running**; memory is copied while it works.
- Downtime is a brief pause during final switchover.
- Fastest with **shared storage** — only RAM moves.
- `qm migrate 101 Host2 --online`

CPU compatibility: A VM using CPU type `host` may fail to live-migrate between unlike CPUs. Use a common model such as `x86-64-v2-AES`.

LIVE DEMO *Live-migrate a running VM with a continuous ping open to it.*

Storage Replication

- **Replication** copies a guest's local ZFS disks to another node on a schedule — every 15 minutes by default.
- After the first full copy, only **incremental** ZFS snapshot changes are sent.
- Lets a guest on local storage move or fail over quickly — the disk is already there.
- Requires **ZFS** on source and target; configure it under **VM → Replication**.



Replication sets your RPO

A 15-minute schedule means up to **15 minutes of changes** can be lost if the source fails. Shorten the interval for critical guests — at the cost of more network traffic.

How the HA Manager Works

Proxmox HA is a manager on each node plus one cluster-wide decision maker:

Piece	Role	Service
Cluster Resource Manager	One active per cluster; decides where each HA resource should run	`pve-ha-crm`
Local Resource Manager	Runs on every node; starts, stops, and monitors its local resources	`pve-ha-lrm`
Watchdog fencing	Self-reboots a node that loses quorum so others can safely take over	`watchdog-mux`
HA resource	A VM or container placed under HA control	`vm:101`, `ct:200`

HA restarts a failed guest on a healthy node — expect a reboot-length outage, not zero downtime.

HA Groups Became Node Affinity Rules

Same idea, new home: “prefer these nodes, in this order.”

	Proxmox VE 8 — HA Groups	Proxmox VE 9 — HA Rules
Where	Datacenter → HA → Groups	Datacenter → HA → Affinity Rules
Node preference	Group members with priorities	Node affinity rule with priorities
Keep apart / together	Not available	Resource affinity rule (positive / negative)
Fail back	`nofailback` set on the group	`failback` set per resource

Lab note: Lab 14 builds HA-Group1. On a 9.x cluster, create the equivalent node affinity rule instead.

Failover Testing

Add a VM to HA, then take its host away and watch the cluster respond:

```
ha-manager add vm:101 --state started --max_restart 1 --max_relocate 1
ha-manager status                  # quorum, master CRM, each LRM, resources
# Power off Host2 from iLO, then watch vm:101 go fence → recovery → started
```

- Expect a delay of about **two minutes**: the lost node must be fenced before its guests restart elsewhere.
- A guest only fails over if its disks are on **shared** or **replicated** storage.
- Use node **maintenance mode** for planned work so guests migrate away cleanly.

LIVE DEMO *Hard power-off Host2 via iLO and narrate ha-manager status as the VM recovers.*

New in Proxmox VE 9.x



Dynamic load balancer (9.2)

The **Cluster Resource Scheduler (CRS)** now watches live CPU and memory use and migrates HA guests to even out busy nodes.

Respects every HA rule. Tuned under **Datacenter → HA → CRS**.



HA arm / disarm (9.2)

Suspends the HA stack **cluster-wide** for a maintenance window, then restores every resource to its prior state.

Prevents surprise fencing during planned work.

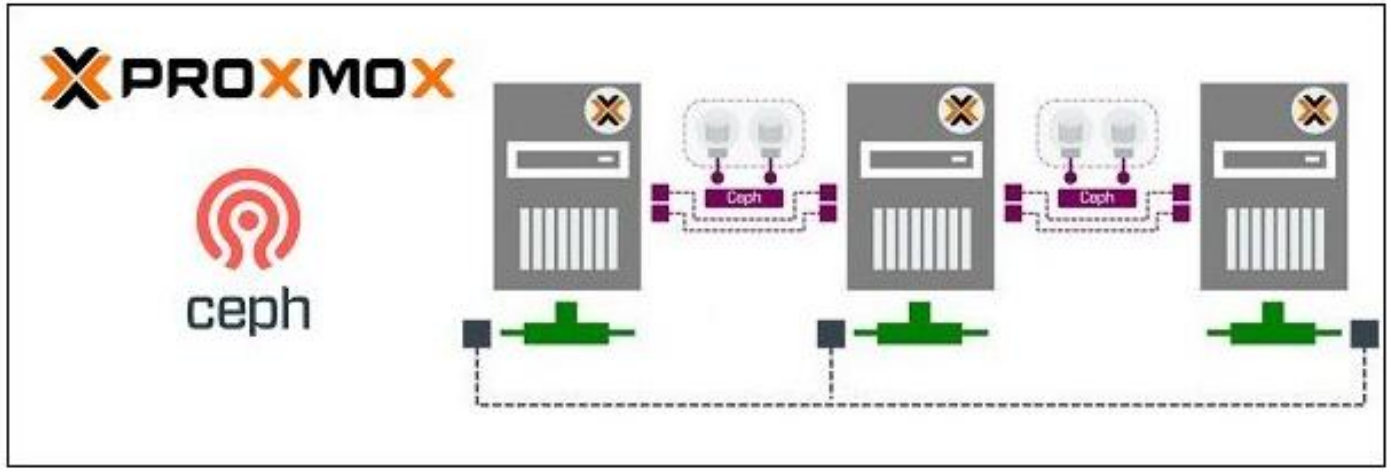
Also in 9.x: HA resource affinity rules (9.0), Debian 13 base, and Ceph Tentacle as the default Ceph release (9.2).

LESSON 3

Ceph Distributed Storage



Hyperconverged Proxmox + Ceph



vs. our lab pods

Pods use a **TrueNAS SAN** over iSCSI/NFS. Ceph replaces that box with the hosts' own drives.

- **Hyperconverged**: every host runs VMs **and** contributes local disks to one shared Ceph pool.
- No external SAN — losing a host loses some copies of data, not access to it.
- Proxmox installs and manages Ceph from the GUI and the `pveceph` tool.

Ceph Daemons

A Ceph cluster is a set of cooperating daemons spread across the hosts:

Daemon	Job	Typical count
Monitor (MON)	Holds the authoritative cluster maps; forms its own quorum	3 (odd number)
Manager (MGR)	Metrics, orchestration, and the dashboard endpoint	2 (active + standby)
OSD	One per disk; stores objects, replicates, and recovers data	One per data disk
Metadata Server (MDS)	Directory and file metadata for CephFS only	1+ if CephFS is used

Modern OSDs use BlueStore, writing directly to the raw device with no file system underneath.

RADOS, CRUSH, and Placement Groups

- **RADOS** (Reliable Autonomic Distributed Object Store) is the object layer everything else is built on.
- Objects are grouped into **placement groups (PGs)**; PGs are mapped to OSDs.
- **CRUSH** computes that mapping — clients calculate where data lives instead of asking a central server.
- When an OSD fails, Ceph **self-heals** by re-replicating its PGs to healthy OSDs.



Failure domains

CRUSH rules place each copy on a different **host** by default — so losing one server never removes every copy of an object.

One Cluster, Three Interfaces

Ceph exposes the same RADOS pool as block, file, or object storage:

Service	Type	Proxmox use
RBD	Block devices (RADOS Block Device)	VM disks and container volumes — the main use
CephFS	POSIX file system (needs MDS)	ISO images, templates, backup targets
RADOS Gateway	S3 / Swift-compatible object store	Not managed by Proxmox; used by apps and backup tools

Snapshots and clones: RBD supports thin provisioning, fast snapshots, and linked clones — the same features you used on ZFS.

Replication vs. Erasure Coding

Two ways to protect data — trading raw capacity against CPU and latency:

	Replicated pool (size 3)	Erasure-coded pool (k=4, m=2)
Raw space per 1 TB	3 TB (33% efficient)	1.5 TB (67% efficient)
Failures survived	2 copies lost (writes pause below min_size 2)	Any 2 chunks lost
Write cost	Low — copy and acknowledge	Higher — compute parity across 6 OSDs
Best fit	VM disks, databases, small random I/O	Archives, backups, large objects

Proxmox defaults new pools to size 3, min_size 2. Do not lower min_size to 1 to "keep things running."

Traditional SAN vs. Ceph

Aspect	Traditional storage	Ceph	Advantage
Scalability	Limited by the array	Add nodes — scales linearly	Growth on demand
Single point of failure	Storage controller	Fully distributed	Reliability
Data protection	RAID within one box	Replication or EC across hosts	Fault tolerance
Cost model	Proprietary hardware	Commodity servers	Lower TCO
Management	Vendor-specific tools	Self-healing, integrated in Proxmox	Less manual work

Fair comparison: At three nodes, a well-run SAN is often simpler. Ceph pays off as the cluster grows.

Deploying Ceph on Proxmox

```
pveceph install --repository no-subscription     # on every node
pveceph init --network 10.0.210.0/24             # public network (example)
pveceph mon create                               # repeat on 3 nodes
pveceph osd create /dev/sdb                      # one per data disk
pveceph pool create vm-pool --add_storages       # size 3 / min_size 2
```

- Current default is **Ceph 20.2 “Tentacle”** (Proxmox VE 9.2); 19.2 “Squid” reached end of support in September 2026.
- Production minimum: **three nodes**, with OSDs on each, and fast dedicated networking.

Network design: Separate the Ceph public and cluster (replication) networks, use 10 GbE or faster, and never share with corosync.

LESSON 4

HA Design and Enterprise Scale

Active-Active vs. Active-Passive

Active-active

- All nodes **serve traffic** at once.
- Better utilization and capacity.
- Needs coordination to keep data consistent.
- Example: web farm behind a load balancer.

Active-passive

- A **standby** takes over only on failure.
- Simpler consistency model.
- Standby capacity sits idle.
- Example: one HA VM restarting on another host.

Choosing: Let the application decide — consistency needs, licensing, and cost weigh more than preference.

RTO, RPO, and the Nines

- **Recovery Time Objective (RTO)**: the longest outage the business accepts.
- **Recovery Point Objective (RPO)**: the most data, measured in time, it can lose.
- HA restart time drives RTO; replication or backup interval drives RPO.

Availability	Down / year	Down / month
99%	3.65 days	7.3 hours
99.9%	8.77 hours	43.8 min
99.99%	52.6 min	4.4 min
99.999%	5.26 min	26 sec

Each additional nine costs roughly ten times more to achieve.

Capacity Planning for Failure

- Size for **N+1**: the cluster must run every HA guest with one node down.
- Three nodes means keeping normal load below about **two-thirds** of total capacity.
- Use history and growth trends — not a single busy day — to forecast.
- Watch Ceph fill: warnings at **85% near-full**; writes stop at **95% full**.



Quick N+1 check

Add up the RAM of all HA guests. If it exceeds the RAM of **two** hosts, a three-node cluster cannot fail over all of them.

Clustering Across Platforms

The concepts carry over; the names change. Preview of Units 13–16:

	Proxmox VE 9	Windows Server 2025 Hyper-V
Membership	Corosync + pmxcfs	Failover Cluster service
Tie-breaker	QDevice	Disk, file share, or cloud witness
Shared storage	Ceph, iSCSI, NFS	CSV, Storage Spaces Direct
Notable new	Dynamic load balancer, affinity rules	Workgroup-cluster live migration, GPU-P live migration

Windows failover clusters support up to 64 nodes and 8,000 VMs per cluster.

Security Across the Cluster



SECURITY FOCUS

Keep corosync and Ceph traffic on **isolated, non-routed networks** — any node in the cluster can rewrite `/etc/pve` for all the others.

Root on one node is effectively **root on every node**. Limit who holds it, and grant roles through cluster-wide permissions.

Never force quorum or lower Ceph `min_size` to 1 to bring services back — both trade a short outage for possible data corruption.

Why it matters: One compromised or misconfigured node can corrupt every VM and configuration file in the cluster at once.

Hands-On Lab 14: Clustering and HA

NETLAB · your three-host pod · dedicated cluster network 10.0.200.x/24



- Create **Cluster1** and join all three hosts on the cluster network.
- Perform **offline** and **live** migrations and record the downtime.
- Configure a **replication** task between hosts.
- Build **HA-Group1**, add a VM as an HA resource, and test automatic failover.

Graded on the live cluster: quorate, migrations completed, and the HA resource recovering after a node failure.

Key Takeaways

- A cluster is one management plane; quorum (a majority of votes) keeps it safe from split-brain.
- Migration moves guests by hand, replication copies them on a schedule, HA restarts them on failure.
- HA depends on fencing and shared or replicated storage — expect minutes of downtime, not zero.
- Ceph turns host disks into shared storage; CRUSH places copies across hosts so failures self-heal.
- Design for failure: set RTO/RPO first, then size the cluster N+1.

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Questions?

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